

## Operations Troubleshooting Reference

Extracted from: 13-operations-troubleshooting.md · Full research repo: [github.com/Crespo1301/conquerorx-research](https://github.com/Crespo1301/conquerorx-research) (private) · Prepared by Carlos Crespo / CSolutions · 2026-08-24

# Operations & Troubleshooting Log

Living record of real ConquerorX incidents observed at Kings Seaport, with diagnosis and resolution. Both a reference (what does this alert mean?) and an incident log (has this happened before? how often?).

## How to use this doc

### When something goes wrong on a shift:

1. Scan the "Known incident patterns" section for a match by symptom.
2. If found, follow the diagnosis + resolution steps.
3. If not found, log it in the "New incidents" section at the bottom with as much detail as you can capture (time, lanes involved, what the guests were doing, exact error text).
4. Later — someone (Carlos + Claude) turns new incidents into new "Known patterns" as we learn what they mean.

### Field entry template — copy into a new incident:

```
### Incident: <one-line summary>
- **When:** YYYY-MM-DD HH:MM ET
- **Lanes / terminals:** e.g. lanes 13-14, front desk terminal 1
- **Symptom:** what did staff / guests see
- **In-progress activity:** who was on those lanes doing what
- **Exact message text (if any):** verbatim
- **Duration:** how long from onset to normal
- **Recovery action:** what fixed it (or "self-recovered")
- **Guest impact:** none / minor / had to comp
```

- **Follow-up:** what still needs to happen (ticket to QubicaAMF, etc.)
- **Diagnosis (added later):** root cause once known

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## The comm chain (authoritative)

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Mapped from `Conqueror-2-386.html` (TCS Overview) and the DLL family under

`Qbk.Lanes.*` .

\_(diagram — see the online repo for a rendered version)\_

## Known incident patterns

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### Pattern A — "No comms" on a pod, both lanes reset together

**Symptom:** two paired lanes (e.g. 13/14, 15/16) show "No Comms" simultaneously

in the front-desk lane grid, then their score consoles reboot on their own.

**What's happening:** the pod's score console lost heartbeat with the

BowlingAgent daemon on the server, hit its watchdog timeout (~5-15 seconds

of missed pings), and rebooted itself as a recovery. This is built-in

QubicaAMF behavior, not a bug — it's the console's "I've lost my brain,

restart" reflex.

### Comm chain that was interrupted:

```
Score console → Q2A protocol → BowlingAgent (server 5130/7014)
                → MxSvc → ConquerorServer → Front-desk UI
```

### Most likely root cause, ranked:

1. **Network glitch on the pod's cable/switch** — Two paired lanes share a

network run; a flap or bump takes both down. Self-recovers when link

returns.

2. **Console hard-locked** — bad game state (weird pinfall pattern) froze the

console. Watchdog kicked in.

3. **Pinsetter fault** — jam trip or overcurrent → pinsetter controller

reboots → score console loses hardware → resets.

4. **Power dip on the pod** — a lone pod power blip.

5. **BowlingAgent hiccup on the server** — GC pause or brief crash. Would affect more pods, so less likely if only 13/14.

6. **Working Copy sync at wrong time** — very rare mid-shift.

#### **Immediate action:**

1. Give it 60-90 seconds — consoles usually reconnect on their own.

2. Verify in the lane grid that 13/14 return to "Ready" or "In Session".

3. Check with guests on the lanes — did their session/scores restore?

(They usually do; scores are persisted server-side.)

4. If TCS (Trouble Call System) fired, dismiss / resolve those tickets.

#### **Follow-up if repeated (> 1 × per week on the same pod):**

- Ask maintenance to inspect the pod's network cable and switch port.
- Ask maintenance to inspect the pinsetter for intermittent faults.
- File a QubicaAMF support ticket including the SystemLog entries around

the incident times.

#### **How to check the logs (needs SQL access):**

- Table: `SystemLog` (in the ConquerorX DB, populated by `qsp_log_insert` )
- Filter by timestamp  $\pm 2$  minutes of the incident
- Look for entries mentioning "lane 13", "lane 14", "heartbeat", "no comms",

"reconnect"

- Also check `LaneHistory` for state transitions (OFF → ON) on 13/14
- `C:\ProgramData\QubicaAMF\BowlingAgent\Logs\` on the server for BA-side

connection logs

**First observed:** 2026-08-24 mid-shift at Kings Seaport (see incident C1

below).

#### **Escalation thresholds:**

| Frequency                   | Response                                  |
|-----------------------------|-------------------------------------------|
| Once a month, self-recovers | Note in shift log, no ticket              |
| Same pod > 1× per week      | Maintenance check on that pod             |
| Multiple pods same day      | Server-side issue — file QubicaAMF ticket |
| Session data lost           | File QubicaAMF ticket immediately         |

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**Pattern B — Terminal shows "Connection to Conqueror server missing"**

**Symptom:** front-desk / back-office terminal can no longer talk to the server. Shows a persistent error dialog or the app just closes.

**Cause:** either MxSvc, ConquerorServer, or the network between terminal and server is down.

**Immediate action:**

1. Wait 30 seconds — brief service restarts happen.
2. Ping the server IP from the terminal (Command Prompt: `ping <server-ip>` ).
3. If ping fails → network/switch issue.
4. If ping works but Conqueror still errors → server-side service issue.

On the server:

- Check Windows Services: `MxSvc` and `MSSQL$CONQUERORX` should both be "Running".
- If not, restart via `RestartServices.exe` (in an elevated PowerShell, from `C:\QDesk\Bin\ConquerorServer\` ).

5. If services restart cleanly but the error persists, restart the server PC — MxSvc has a startup dependency on SQL Server that can get out of order.

**Never known to happen without a root cause.** If it happens, log it.

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## Pattern C — Cloud connection lost

**Symptom:** notification like "Conqueror Server cannot connect to the Cloud" or "Cloud is temporarily not available".

**What still works locally:** everything — reservations, POS, lane control, payments. The cloud is only for centralization sync, plugin loading, and updates.

**What breaks:** new customer signups (if configured to require cloud validation), cross-center loyalty lookups, CloudPlugin loading, Working Copy sync of new versions.

### Immediate action:

- **Wait 5 minutes** — most cloud interruptions are QubicaAMF-side and auto-recover.
- Check internet from the server PC (open a browser, load any external site).
- Try `QCcloudTestConnection.exe` from `C:\QDesk\Bin\ConquerorServer\` — it runs a diagnostic to `qcloud.qubicaamf.com`.

**If persistent > 30 minutes:** call QubicaAMF support. Local operations can continue in the meantime.

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## Pattern D — Reservation import rejected

**Symptom:** the reservation `.xls` upload fails with a "not valid" or "please specify ..." error message.

**Cause:** violates the ConquerorX reservation import contract. All known error messages and their fixes are documented in

[ `08-templates-and-imports.md` ]([08-templates-and-imports.md#documented-import-error-messages-from-the-en-translation-file](#))

and the reservations-builder `CHANGELOG.md` .

**Immediate action:** cross-reference the error verbatim to the fixes list.

Common ones:

| Error                                                                      | Fix                                                                        |
|----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| "The start date specified in the Excel file is not valid"                  | Column 0 must be an Excel date serial; also date must be today or future   |
| "The lane type specified in the Excel file is not valid"                   | Column 3 must be <code>1</code> (Single) or <code>2</code> (Pair)          |
| "Please specify the reservation name in the Excel file"                    | Column 2 must have a non-blank string                                      |
| "Not all the genders specified in the Excel file are valid"                | Bowler gender column must be <code>1</code> or <code>2</code> (0 rejected) |
| "Unable to import reservations in the past"                                | Date is past-dated — regenerate for today or later                         |
| "Unable to import reservation because there are not enough bookable lanes" | Assigned lanes don't exist or aren't bookable in ConquerorX for that time  |

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## New incidents

Add new incidents here as they happen. Move them into a "Pattern" section above once we understand them well enough to write a diagnosis.

### C1 — Lanes 13/14 reset with "no comms" then rebooted

- **When:** 2026-08-24, mid-shift (exact time not captured)
- **Lanes / terminals:** lanes 13, 14
- **Symptom:** lane grid showed "No Comms" on both, then their consoles

spontaneously started rebooting

- **In-progress activity:** unknown — mid-shift, likely a party bowling
- **Exact message text:** "no comms" (paraphrased from staff report)
- **Duration:** unknown; recovery observed
- **Recovery action:** self-recovered (consoles rebooted, reconnected)
- **Guest impact:** unknown (session state on those lanes not verified in

the moment)

- **Follow-up:** watch for repeats on this pod; add to shift log for

pattern tracking

- **Diagnosis:** classic Pattern A above. Most likely a brief network glitch

on the 13/14 pod or a console watchdog reset. First occurrence — no escalation needed yet.

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## TCS (Trouble Call System) — how QubicaAMF designed it

Authoritative content from `Conqueror-2-385.html` through `Conqueror-2-401.html` .

**BES, Bowland, and Bowland-X only.** Older scoring generations don't run

TCS.

### The four actors that raise a trouble call

Per `Conqueror-2-388.html` "Signaling Errors":

| Actor                  | How they raise a call                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MAG 3 Pinsetter</b> | Intelligent auto-detection. The pinsetter identifies its own problems and raises the alarm directly, cutting out any human. Fastest path.                                                                 |
| <b>Bowler</b>          | Presses button on the ball return, OR (on BES lanes) uses the lane console menu. Requires "Mechanic Call" option enabled in Setup → Bowling Setup → Lane Options. Default message: "Lane N, Bowler call". |
| <b>Operator</b>        | From Lane Status: Mechanic Service → Mechanic Call, enter message, OK. OR from Back Office → TCS → New, select lane, enter voice message, OK. Default message if none entered.                            |
| <b>Mechanic</b>        | Doesn't raise, but resolves. Acknowledges via button behind the pinsetter OR via Acknowledge on the telephone. Puts lane into Work in Progress mode.                                                      |

### Alarm surfaces that fire

Per `Conqueror-2-389.html` "Alarms and User Warnings":

- **Telephone call** to the mechanic with synthesized voice message describing the error. Repeats until response.
- **Loud-speaker voice message** in the center. Repeats until response.
- **Alarm siren.**
- **Front-desk pop-up:** "Intervention required on lane N" (if pinsetter- or bowler-initiated; toggleable in Setup → Terminal Setup → Preferences).
- **Per-lane light** behind each lane, flashing.
- **General light** above the lanes, indicating an out-of-order lane.

All of the above continue firing until acknowledgement or cancellation.

## Telephone command codes (yes, the mechanic uses a phone)

From `Conqueror-2-401.html` — touch-tone commands the mechanic enters after picking up the auto-call:

| Key | Command                         |
|-----|---------------------------------|
| 1   | Acknowledge                     |
| 2   | Repairs Completed               |
| 3   | Cancel Request                  |
| 4   | Perform a Pinsetter Partial Set |
| 5   | Perform a Pinsetter Full Set    |
| 6   | Spot Pins                       |
| 9   | Record Voice Message            |
| #   | Abort a Command                 |

## TCS privileges

Per `Conqueror-2-398.html` — permissions granted per staff role:

- Make a New Trouble Call
- Acknowledge a Call
- Complete a Call
- Cancel a Call
- Access TCS Plugin
- Access TCS Setup Plugin

## TCS reports (7 built-in report templates)

From `Conqueror-2-393.html` :

- `TCS.rpt` — master report
- `TCSDownTime.rpt` — lane downtime aggregate
- `TCSErrorPerCenter.rpt` — errors per center (chain-wide view)
- `TCSErrorPerLane.rpt` — errors per lane (which pod fails most)
- `TCSTypeOfErrorPerCenter.rpt` — error-type distribution
- `TCSWorkshop.rpt` — mechanic workshop activity
- `TcsVocalMessages.rpt` — voice-message log



For Kings' opening/closing managers: **TCSErrorPerLane.rpt** is the report that flags recurring hardware problems (Pattern A escalation from above).

### TCS setup — records retention

Per `Conqueror-2-396.html` "Alarm Checks":

- **Keep Records for \_ Days** — configurable retention. Center chooses how long TCS history is kept.

### Escalation contacts

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Fill in with real Kings-side and QubicaAMF-side contact info once known:

| Situation                       | Contact                                            |
|---------------------------------|----------------------------------------------------|
| Local network / cable issue     | Kings maintenance team                             |
| Pinsetter / lane hardware fault | Kings mechanical team                              |
| Persistent multi-pod outage     | QubicaAMF support                                  |
| Data-loss incident              | QubicaAMF support (file with logs)                 |
| Software bug / crash            | QubicaAMF support (build TechSupport bundle first) |

### Building a TechSupport bundle for QubicaAMF

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When escalating anything to QubicaAMF:

1. Run `C:\QDesk\Bin\TechSupportClient.exe` (client) or `C:\QDesk\Bin\ConquerorServer\TechSupport.exe` (server) — one of them packages logs + config + a DB snapshot into a zip.
2. Note the incident time in local timezone (QubicaAMF is in Europe / East Coast, be explicit).
3. Attach the bundle to the support ticket.

### Reference

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- Comm chain and root-cause reasoning: [ 03-services-and-processes.md ](03-services-and-processes.md)
- Log source ( qsp\_log\_insert → SystemLog ): [ 05-database-schema.md ](05-database-schema.md)
- BowlingAgent details: [ 03-services-and-processes.md ](03-services-and-processes.md#running-processes-owned-by-conquerorx-server-role)